

Score: / 10

Calculus I (MATH 2200)
Spring 2020
Wednesday, 02/12/2020
Quiz 3: Section 2.5, 2.6

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Section 2.5

1. Determine the *end behavior* (that is, the limit as $x \rightarrow \pm\infty$) of the following functions:

1 pts

(a) $f(x) = \frac{3x + 2}{x^2 - 1}$

As $f(x)$ is a rational function, and as the degree of the polynomial in the denominator of f is larger than the degree of the polynomial in the numerator, the limit as x goes to $\pm\infty$ is zero. Correspondingly, $y = 0$ is a horizontal asymptote of $f(x)$.

1 pts

(b) $g(x) = \frac{40x^4 + 4x^2 - 1}{10x^4 + 8x^2 + 1}$

As $f(x)$ is a rational function, and as the degree of the polynomial in the denominator of f is the same as the degree of the polynomial in the numerator, the limit as x goes to $\pm\infty$ is equal to the ratio of the leading terms. Correspondingly, $y = \frac{40}{10} = 4$ is a horizontal asymptote of $f(x)$.

2 pts

(c) $g(x) = \frac{91x^7 - x^2 + 2}{1000x^3 + 800x^2}$

As $f(x)$ is a rational function, and as the degree of the polynomial in the denominator of f is smaller than the degree of the polynomial in the numerator, $f(x)$ becomes arbitrarily large as x goes to $\pm\infty$. Correspondingly, as the leading terms of both the numerator and denominator are positive, $f(x)$ becomes large and positive as x approaches ∞ . Hence, $\lim_{x \rightarrow \infty} f(x) = \infty$. Similarly, as the denominator is a polynomial of odd degree and the numerator is a polynomial of odd degree, $f(x)$ becomes large and positive as x approaches $-\infty$. Hence, $\lim_{x \rightarrow -\infty} f(x) = \infty$.

2 pts

2. Determine the end behavior of $f(x) = \frac{10x^2 + x + 1}{x^{2.2} + \sqrt{x}}$.

$$\begin{aligned}\lim_{x \rightarrow \infty} f(x) &= \lim_{x \rightarrow \infty} \frac{10x^2 + x + 1}{x^{2.2} + \sqrt{x}} \\&= \lim_{x \rightarrow \infty} \frac{10x^2 + x + 1}{x^{2.2} + x^{0.5}} \\&= \lim_{x \rightarrow \infty} \left(\frac{x^{-2.2}}{x^{-2.2}} \right) \frac{10x^2 + x + 1}{x^{2.2} + x^{0.5}} \\&= \lim_{x \rightarrow \infty} \frac{10x^2(x^{-2.2}) + x(x^{-2.2}) + (x^{-2.2})}{x^{2.2}(x^{-2.2}) + x^{0.5}(x^{-2.2})} \\&= \lim_{x \rightarrow \infty} \frac{\frac{10}{x^{0.2}} + \frac{1}{x^{1.2}} + \frac{1}{x^{2.2}}}{1 + \frac{1}{x^{1.7}}} \\&= \frac{0 + 0 + 0}{1 + 0} = 0\end{aligned}$$

Similarly, we have $\lim_{x \rightarrow -\infty} f(x) = 0$ as well, which yields that f has $y = 0$ as a horizontal asymptote.

Section 2.6

- 2 pts 1. Let $f(x) = \begin{cases} \frac{x^2 + x - 6}{x - 2} & \text{if } x \neq 2 \\ 7 & \text{if } x = 2 \end{cases}$. Is $f(x)$ continuous at $x = 2$? Explain.

For a function to be continuous at a point, the limit must exist at that point, and it must equal the value of the function at that point. If the limit exists at a point, it must match up with the left-hand and right-hand limits at that point. First, we calculate the left-hand limit:

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} \frac{x^2 + x - 6}{x - 2} = \lim_{x \rightarrow 2^-} \frac{(x - 2)(x + 3)}{x - 2} = \lim_{x \rightarrow 2^-} x + 3 = 5.$$

Then, we calculate the right-hand limit:

$$\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} \frac{x^2 + x - 6}{x - 2} = \lim_{x \rightarrow 2^+} \frac{(x - 2)(x + 3)}{x - 2} = \lim_{x \rightarrow 2^+} x + 3 = 5.$$

As these are equal, we then yield that

$$\lim_{x \rightarrow 2} f(x) = 5.$$

However, $f(2) = 7$, so while the limit does exist at $x = 2$, the limit does not equal the value of the function at $x = 2$. Thus, $f(x)$ is not continuous at $x = 2$.

2 pts

2. Evaluate the limit $\lim_{x \rightarrow 2} \cos\left(\frac{x^2 - 4}{x - 2}\right)$.

First, recall that $\cos(x)$ is continuous at all points of its domain. Setting $f(x) = \cos(x)$ and $g(x) = \frac{x^2 - 4}{x - 2}$, we then have $\cos\left(\frac{x^2 - 4}{x - 2}\right) = (f \circ g)(x)$. Now, let us notice that $g(x)$ is not continuous at 2, as $g(x)$ is not defined at 2. However, by the properties of continuous functions, we can still evaluate this limit provided that $\lim_{x \rightarrow 2} g(x)$ exists. So, we calculate:

$$\lim_{x \rightarrow 2} g(x) = \lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} = \lim_{x \rightarrow 2} \frac{(x - 2)(x + 2)}{x - 2} = \lim_{x \rightarrow 2} x + 2 = 4.$$

So, we then may see that

$$\lim_{x \rightarrow 2} \cos\left(\frac{x^2 - 4}{x - 2}\right) = \lim_{x \rightarrow 2} (f \circ g)(x) = f\left(\lim_{x \rightarrow 2} g(x)\right) = \cos(4) \approx -0.654.$$